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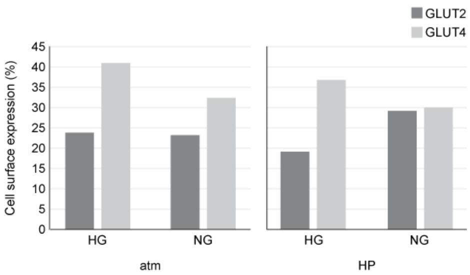

Figure 1 Cell surface expression of GLUT transporters in response to mechanical stress and varying glucose concentration

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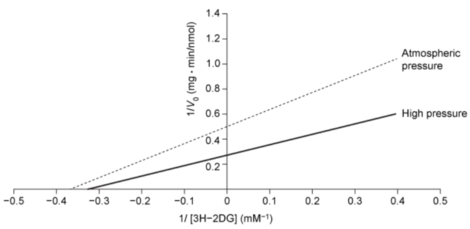


Figure 2 Plot of glucose uptake in podocytes with and without mechanical stress

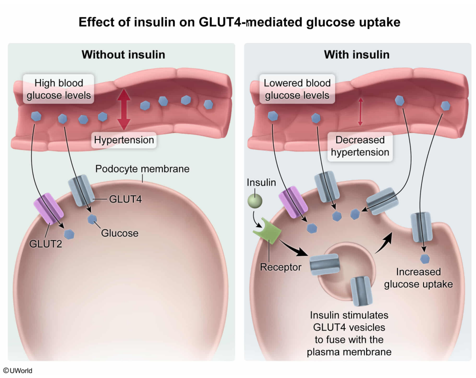
Which mechanism best describes how insulin could alleviate capillary hypertension in diabetic patients?

- A. Insulin increases the concentration of glucose inside cells that express GLUT2. [3%]
- B. Insulin decreases blood glucose concentration by stimulating GLUT4. [92%]
- C. Insulin increases blood glucose concentration by inhibiting glucokinase. [1%]
- D. Insulin decreases glucose uptake by increasing glucose-6-phosphate levels. [2%]

Correct
91% answered correctly

Explanation:

Explanation:



According to the passage, capillary hypertension (high blood pressure) increases the mechanical stress on podocytes. In diabetic patients, high blood glucose levels contribute to rising blood pressure in capillaries. Insulin administration **reduces** the concentration of glucose in the blood by **increasing glucose uptake** and consumption.

Cells obtain glucose through transporters such as GLUT4 and GLUT2; however, **only GLUT4** is susceptible to regulation by **insulin**. GLUT4 is an inducible transporter that responds to the presence of insulin by increasing its uptake of glucose. Consequently, blood glucose concentrations are lower and capillary hypertension decreases when insulin binding stimulates GLUT4.

(Choice A) GLUT2 is not susceptible to insulin regulation. Therefore, the concentration of glucose inside cells with GLUT2 receptors is not altered by insulin.

(Choices C and D) Glucokinase is an isozyme of hexokinase, the enzyme that converts glucose into glucose-6-phosphate in the first step of glycolysis. The release of insulin stimulates glucokinase and results in an increase in glucose-6-phosphate production that consumes the glucose within the cell. As a result, glucose uptake *increases* and blood glucose levels *decrease*.

Educational objective:
Insulin reduces blood glucose concentrations by increasing the rate of glucose uptake and glucose consumption. Glucokinase converts glucose into glucose-6-phosphate in the first step of glycolysis. Insulin release stimulates glucokinase and increases the production of glucose-6-phosphate.

Time spent:0 sec	**QID**:402046
Subject: Biochemistry	**System:** Enzymes

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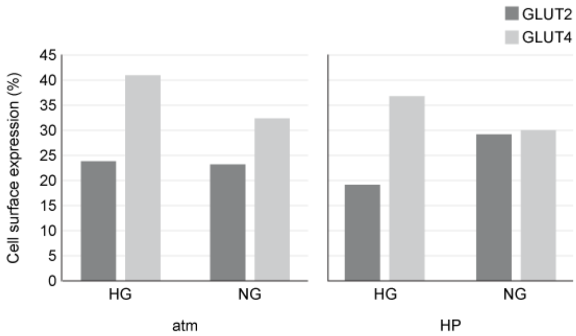

Figure 1 Cell surface expression of GLUT transporters in response to mechanical stress and varying glucose concentration

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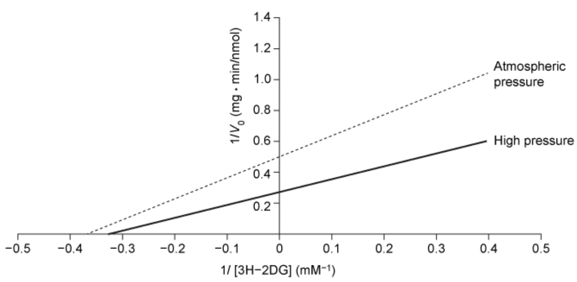


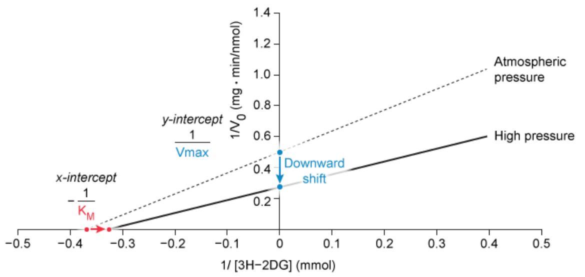
Figure 2 Plot of glucose uptake in podocytes with and without mechanical stress

What stress-induced change in Figure 2 provides evidence that mechanical stress alters the binding properties of the GLUT transporters?

- A. The left shift of $-1/K_M$ [18%]
- B. The downward shift of $1/V_{max}$ [28%]
- C. The decreased catalytic efficiency [10%]
- D. The lower affinity for the substrate [42%]

Correct
41% answered correctly

Explanation:



Lineweaver-Burk plots can be used to compare the effects of different experimental conditions on kinetic parameters. The linear graphs are derived from the Michaelis-Menten equation and display the inverse reaction velocity ($1/V_0$) versus the inverse substrate concentration (1/[S]). Consequently, $\boldsymbol{K_M}$ is the **reciprocal** of the negative ***x*-intercept**, and V_{max} is the reciprocal of the *y*-intercept.

Figure 2 shows a Lineweaver-Burk plot of the inverse rate of glucose uptake as a function of inverse 3H-2DG concentration under both atmospheric (atm) and high (HP) air pressure. The question states that mechanical stress alters the binding properties of the GLUT transporters. The binding affinity of a protein for its substrate is exhibited by the K_M value, for which **lower** values indicate **higher** binding affinities. Therefore, changes in the *x*-intercept should be evident in podocytes exposed to HP. In comparison to atmospheric air pressure, high pressure conditions cause a right shift in the *x*-intercept. A right shift shows that the K_M value has **increased**, which suggests a lower binding affinity for the substrate.

(Choice A) A left shift in the *x*-intercept on a Lineweaver-Burk plot indicates that the K_M value has *decreased*. Stress generated by high pressure in Figure 2 resulted in a *right shift* of $-1/K_M$, indicating an *increased* K_M value.

(Choice B) Changes in the binding affinity of a protein do not alter the value of V_{max}. A downward shift in a Lineweaver-Burk plot corresponds to an *increase* in V_{max}, which suggests that glucose uptake is faster under mechanical stress.

(Choice C) Catalytic efficiency is the k_{cat}/K_M ratio of an enzyme and is inversely proportional to the slope of a Lineweaver-Burk plot. In Figure 2, the slope of the plot decreases, indicating an *increase* in catalytic efficiency.

Educational objective:
Changes in the binding properties of a protein can alter the K_M value. On a Lineweaver-Burk plot, K_M is the negative reciprocal of the *x*-intercept, and V_{max} is the reciprocal of the *y*-intercept. A right shift of the *x*-intercept corresponds to an increase in K_M whereas a left shift indicates a decrease.

Time spent: 0 sec	**QID**:402047
Subject: Biochemistry	**System:** Enzymes

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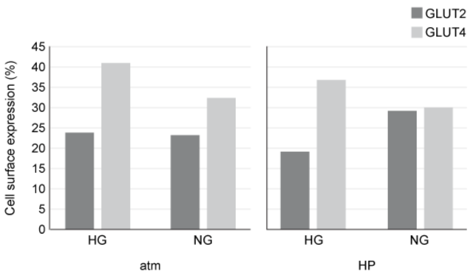

Figure 1 Cell surface expression of GLUT transporters in response to mechanical stress and varying glucose concentration

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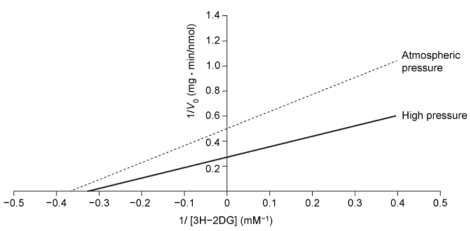


Figure 2 Plot of glucose uptake in podocytes with and without mechanical stress

Based on the data in Figure 1, which experimental conditions will generate the highest maximum velocity for each transporter?

○ A.

GLUT 2	GLUT 4
atm	HP
HG	NG

[4%]

○ B.

B.

GLUT 2	GLUT 4
HP	HP
HG	HG

[3%]

C.

GLUT 2	GLUT 4
HP	atm
NG	HG

[90%]

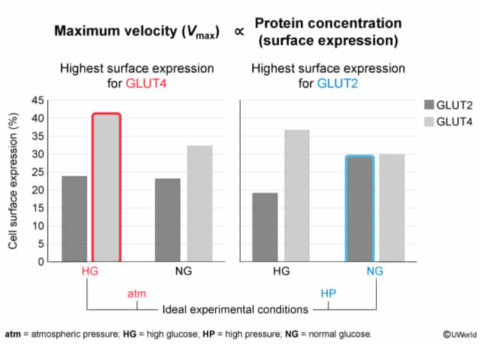
D.

GLUT 2	GLUT 4
atm	atm
NG	NG

[1%]

Correct
89% answered correctly

Explanation:



Glucose in the glomerular capillaries is removed by the GLUT transporters on the **cell membrane** of podocytes. In the experiment described, the amount of radiolabeled glucose (3H-2DG) transported was **directly proportional** to the cell surface expression of GLUT transporters. Therefore, the **maximal velocity** V_{max} of glucose uptake was dependent on the **expression** (ie, concentration) of GLUT2 and GLUT4 transporters on the cell membrane. Researchers tested the effect of mechanical stress and hyperglycemia on the expression of GLUT transporters by varying air pressure and 3H-2DG concentration.

The bar graph in Figure 1 illustrates the surface expression of GLUT2 and GLUT4 in all the possible combinations of experimental conditions. The bar with the **highest surface expression** represents the set of experimental conditions that generated the **highest** $\boldsymbol{V_{max}}$. For GLUT2, the highest expression was observed in cells maintained at high (HP) air pressure and under normal glucose (NG) concentrations. In contrast, GLUT4 expression levels were highest at atmospheric (atm) pressure and hyperglycemic (HG) conditions. Of the choices presented, only the parameters given in Choice C correctly identify these ideal experimental conditions for GLUT2 and GLUT4.

(Choice A) The optimal experimental conditions for GLUT2 and GLUT4 are switched here.

(Choice B) Neither GLUT2 nor GLUT4 obtains its highest expression levels when exposed to high air pressure and hyperglycemic conditions.

(Choice D) Atmospheric pressure increased GLUT4 expression but not GLUT2 expression. Normal glucose concentrations increased GLUT2 expression but not GLUT4. GLUT4 requires hyperglycemic glucose concentrations, and GLUT2 expression is highest under high air pressure.

Educational objective:
The maximum reaction velocity V_{max} is directly proportional to the enzyme concentration. The concentration of a protein or enzyme in a cell is also referred to as its expression level.

Time spent: 0 sec	**QID**:402048
Subject:	**System:**
Biochemistry	Enzymes

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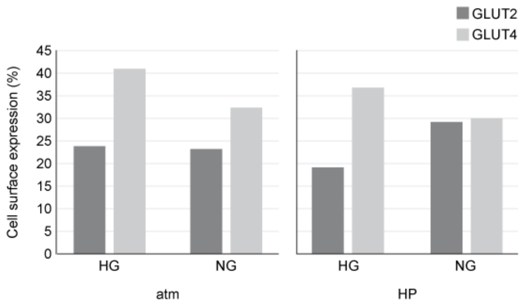


Figure 1 Cell surface expression of GLUT transporters in response to mechanical stress and varying glucose concentration

In vivo studies have shown that transport via GLUT2 is not inducible by insulin and depends only on extracellular glucose concentrations. However, GLUT4-mediated uptake is induced by insulin binding and occurs under high glucose conditions when insulin is released, leading to trafficking of GLUT4 to the cell surface. The standard K_M values for GLUT2 and GLUT4 are 19.4 mM and 4.3 mM, respectively, with normal blood glucose levels after a meal being ~7.8 mM. The data shown in Figure 2 represent the combined kinetic properties of both transporters.

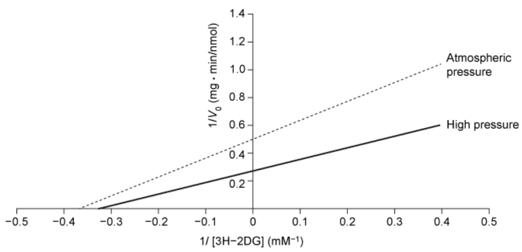


Figure 2 Plot of glucose uptake in podocytes with and without mechanical stress

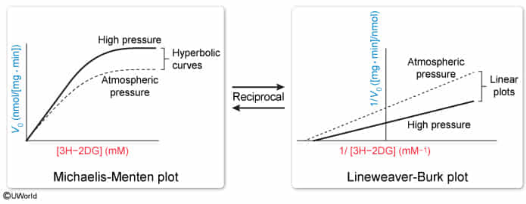
Researchers inverted the data shown in Figure 2 and observed that the values fit a hyperbolic curve. What conclusion is best supported by their observation?

◯ A. V_0 was measured after equilibrium was achieved. [3%]

- [x] B. Glucose uptake followed Michaelis-Menten kinetics. [75%]
- [ ] C. GLUT transporters exhibited cooperativity as [3H-2DG] rose. [12%]
- [ ] D. The concentration of [3H-2DG] was the limiting factor for V_{max}. [8%]

Correct
76% answered correctly

Explanation:



Lineweaver-Burk plots are double-reciprocal graphs that invert the reaction rate V_0 and substrate concentration [S] of the Michaelis-Menten equation to produce linear graphs; therefore, the *x*- and *y*-axes measure 1/[S] and $1/V_0$, respectively. Inverting the data of a Lineweaver-Burk plot regenerates the Michaelis-Menten plot, which displays the reaction rate as a function of substrate concentration. In this plot, the reaction rate and substrate concentration maintain a nearly linear relationship when the substrate concentration is low. However, as more substrate is added to the reaction, active sites become saturated and the reaction rate plateaus. Graphically, the combination of these two phases produces a **hyperbolic curve** that is characteristic of Michaelis-Menten kinetics.

Figure 2 is a Lineweaver-Burk plot derived from a Michaelis-Menten plot, so inverting the data regenerates the hyperbolic curve observed by the researchers. The most likely explanation for this distinctive shape is that **glucose uptake followed Michaelis-Menten kinetics**.

(Choice A) After equilibrium is achieved, V_0 can no longer be measured as the reaction does not proceed any further. When equilibrium is achieved, the net reaction rate becomes zero.

(Choice C) The presence of cooperativity during glucose uptake would be established by a sigmoidal curve rather than a hyperbolic curve.

(Choice D) The limiting factor for V_{max} in Michaelis-Menten reactions is the enzyme concentration, not the substrate concentration.

Educational objective:
Reactions that follow Michaelis-Menten kinetics produce a hyperbolic curve that shows reaction rate V_0 as a function of substrate concentration [S]. A Lineweaver-Burk plot is a double-reciprocal plot of the Michaelis-Menten equation and shows a linear relationship between $1/V_0$ and 1/[S]. The Lineweaver-Burk plot can be inverted to regenerate the hyperbolic curve.

Time spent: 0 sec	**QID:** 402049
Subject: Biochemistry	**System:** Enzymes

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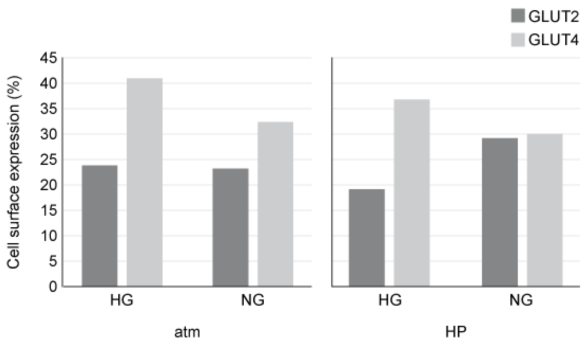

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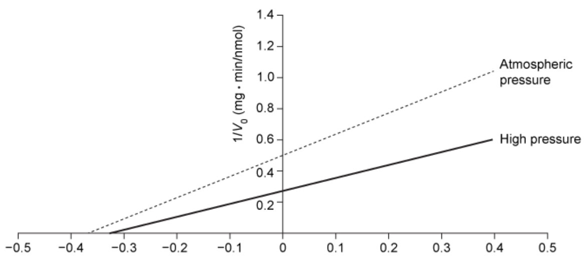


1/ [3H−2DG] (mM⁻¹)

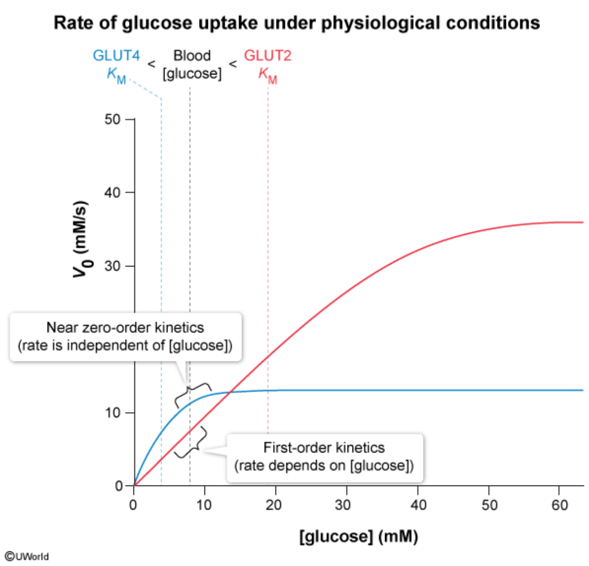
Figure 2 Plot of glucose uptake in podocytes with and without mechanical stress

Which statement about glucose transport under physiological conditions is consistent with the kinetic data provided in the passage?

- A. GLUT2-mediated uptake resembles a first-order process. [54%]
- B. ATP is required for glucose uptake by both GLUT transporters. [7%]
- C. In the fed state, glucose concentration limits GLUT4-mediated uptake. [29%]
- D. Glucose transport is independent of the concentration gradient. [8%]

Correct
55% answered correctly

Explanation:



Rate laws are experimentally determined equations that express a reaction rate in terms of the concentrations of reactants. Most simple biochemical processes and enzymatic reactions have rate laws that can be classified as either zero-order or first-order. For **zero-order reactions**, the rate is **independent** of the substrate concentration. Biochemically, these reactions can occur when there is an excess of substrate and a limited amount of enzyme. In such a case, the rate is only dependent on the enzyme concentration and activity, represented by the rate constant k_{cat}. In contrast, the reaction rate of a **first-order process** *is dependent on substrate* concentration.

The $\boldsymbol{K_M}$ value of an enzymatic reaction represents the substrate concentration at which half the active sites of an enzyme population are filled, so the K_M value can be used to predict the apparent reaction order under various conditions. When K_M is greater than the substrate concentration, the majority of active sites in the enzyme are available and

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the reaction behaves like a first-order process dependent on substrate concentration. However, if the K_M value is substantially less than the substrate concentration, then most of the active sites are saturated and the reaction resembles a zero-order process.

According to the passage, GLUT transporters under physiological conditions are exposed to a glucose concentration of approximately 7.8 mM after a meal. The passage states that GLUT2 transport is dependent on extracellular glucose concentration, and the K_M value of 19.4 mM is much greater than 7.8 mM. Therefore, under these conditions, GLUT2-mediated uptake resembles a **first-order process**.

(Choice B) Glucose uptake occurs via passive transport, which does not require ATP.

(Choice C) According to the passage, the normal glucose concentration in the fed state is ~7.8 mM. This value is substantially greater than the K_M value (4.3 mM) of GLUT4, which indicates that under these conditions GLUT4 is approaching saturation and glucose concentration is not the limiting factor.

(Choice D) Glucose enters cells *down* a concentration gradient through the passive GLUT transporters.

Educational objective:
Enzymatic reactions typically behave as zero-order or first-order reactions. In zero-order reactions, the rate is only dependent on the rate constant k_{cat} because substrate concentrations exceed the K_M value. First-order reactions depend on substrate concentration and occur when K_M is greater than the substrate concentration.

Time spent:0 sec
Subject:
Biochemistry

QID:402050
System:
Enzymes

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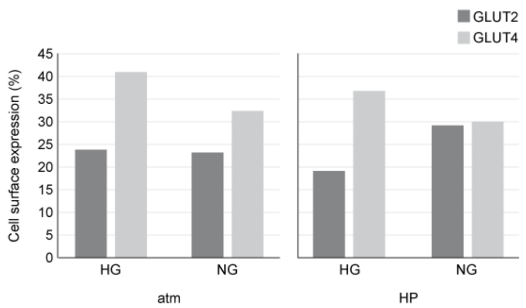

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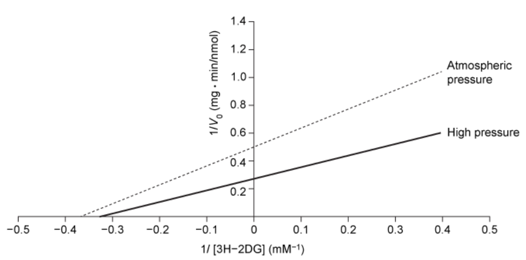


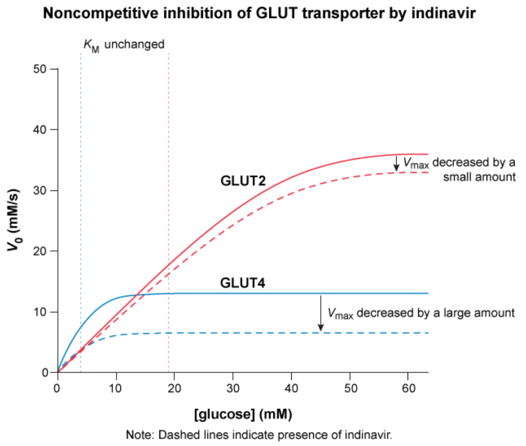
Figure 2 Plot of glucose uptake in podocytes with and without mechanical stress

The drug indinavir inhibits GLUT isoforms noncompetitively. Researchers predict that GLUT4 is significantly more susceptible to indinavir than GLUT2. Which of the following observations would best support their hypothesis?

- ○ A. Indinavir binds the GLUT4-glucose complex exclusively. [6%]
- ○ B. GLUT2 transports less glucose than GLUT4 does when indinavir is bound. [2%]
- ○ C. GLUT2 has a lower K_M value than GLUT4 in the presence of indinavir. [14%]
- ✔ ● D. Indinavir decreases the V_{max} of GLUT4 more than the V_{max} of GLUT2. [76%]

Correct
76% answered correctly

Explanation:



Inhibitors are compounds that alter protein activity by changing the reaction rate V_{max} or the binding affinity K_M of the protein. Based on the information in the question, GLUT transporters are susceptible to **noncompetitive inhibition** by the drug indinavir. Noncompetitive inhibitors bind to regions other than the active site and have **equal affinity** for the **free protein** and the **protein-substrate complex**. As a result, noncompetitive inhibition causes a **decrease in V_{max}** but has **no effect on K_M**.

Although indinavir inhibits both GLUT transporters, researchers predict that indinavir has a greater impact on GLUT4 than on GLUT2. In the presence of the inhibitor, GLUT4 exhibits a **lower V_{max}** in comparison to GLUT2. Therefore, indinavir decreases the V_{max} of GLUT4 to a greater extent than it does the V_{max} of GLUT2.

(Choice A) Indinavir is noncompetitive and binds to *both* the free GLUT4 transporter and the GLUT4-glucose complex. Only uncompetitive inhibitors bind exclusively to the protein-substrate complex.

(Choice B) The inhibitor has a greater effect on GLUT4-mediated glucose uptake than on GLUT2. More glucose should be transported by GLUT2 than by GLUT4.

(Choice C) Noncompetitive inhibition does not affect the K_M value of an enzymatic reaction, so neither transporter has a decreased K_M.

Educational objective:
Noncompetitive inhibitors bind enzymes at sites other than the active site, binding the free enzyme and the enzyme-substrate complex with equal affinity. This type of inhibition is characterized by a decrease in V_{max} with no change in the K_M value.

Time spent:0 sec
Subject:
Biochemistry

QID:402051
System:
Enzymes